

CS 2601 Linear and Convex Optimization

5. Convex optimization problems (part 2)

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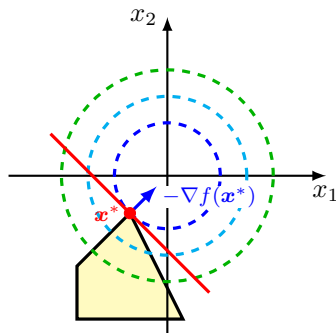
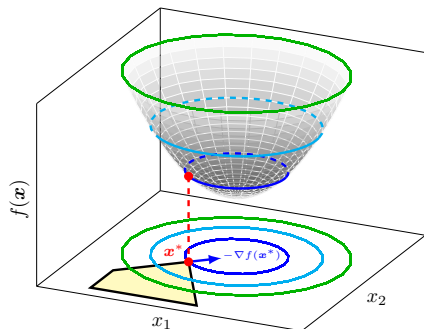
Outline

- Quadratic program and quadratically constrained QP
- Geometric program

Quadratic program (QP)

$$\begin{aligned} \min_x \quad & \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{B} \mathbf{x} \leq \mathbf{d} \\ & \mathbf{A} \mathbf{x} = \mathbf{b} \end{aligned}$$

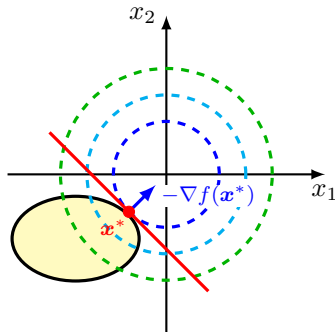
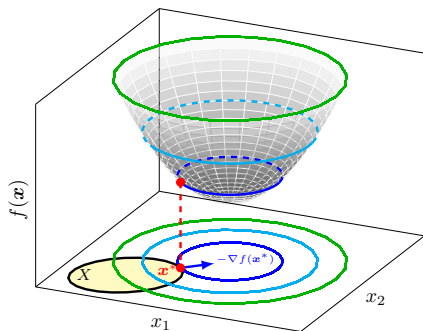
QP is convex iff $\mathbf{Q} \succeq \mathbf{0}$. Reduces to LP if $\mathbf{Q} = \mathbf{0}$.



Quadratically constrained quadratic program (QCQP)

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \frac{1}{2} \mathbf{x}^T \mathbf{Q}_i \mathbf{x} + \mathbf{c}_i^T \mathbf{x} + d_i \leq 0, \quad i = 1, 2, \dots, m \\ & \mathbf{A} \mathbf{x} = \mathbf{b} \end{aligned}$$

QCQP is convex if $\mathbf{Q} \succeq \mathbf{0}$ and $\mathbf{Q}_i \succeq \mathbf{0}, \forall i$. Reduces to QP if $\mathbf{Q}_i = \mathbf{0}, \forall i$.



Example: Linear least squares regression

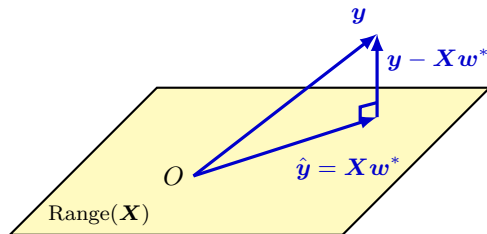
Given $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times p}$, find $\mathbf{w} \in \mathbb{R}^p$ s.t.

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$$

- convex QP with objective

$$f(\mathbf{w}) = \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{y}^T \mathbf{X} \mathbf{w} + \mathbf{y}^T \mathbf{y}$$

Geometrically, we are looking for the orthogonal projection $\hat{\mathbf{y}}$ of $\mathbf{y}$ onto $\text{Range}(\mathbf{X})$, the column space of $\mathbf{X}$. Does the solution always exist?



Example: Linear least squares regression (cont'd)

By the first-order optimality condition, $\mathbf{w}^*$ is optimal iff

$$\nabla f(\mathbf{w}^*) = \mathbf{0}$$

i.e. $\mathbf{w}^*$ is a solution of the **normal equation**,

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\mathbf{w}) = \mathbf{0} \iff \mathbf{X}^T\mathbf{X}\mathbf{w} = \mathbf{X}^T\mathbf{y}$$

Note. $\mathbf{X}^T(\mathbf{y} - \mathbf{X}\mathbf{w}^*) = \mathbf{0}$ means precisely $\mathbf{y} - \mathbf{X}\mathbf{w}^* \perp \text{Range}(\mathbf{X})$.

Case I. $\mathbf{X}$ has full column rank, i.e. $\text{rank } \mathbf{X} = p$

- $\mathbf{X}^T\mathbf{X} \succ \mathbf{0}$
- unique solution

$$\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Case II. $\mathbf{X}$ does not have full column rank, i.e. $\text{rank } \mathbf{X} < p$.

- $\mathbf{X}^T\mathbf{X}$ is singular
- infinitely many solutions

Example: Linear least squares regression (cont'd)

Example. Solve

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$$

with

$$\mathbf{X} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}.$$

Solution. The normal equation is

$$\mathbf{X}^T \mathbf{X} \mathbf{w} = \mathbf{X}^T \mathbf{y}$$

with

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{X}^T \mathbf{y} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

Since $\mathbf{X}$ has full column rank,

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \begin{bmatrix} 1.5 \\ 2 \end{bmatrix}$$

Example: Linear least squares regression (cont'd)

Example. Solve

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$$

with

$$\mathbf{X} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}.$$

Solution. Note $\text{rank } \mathbf{X} = 2 < 3$. The normal equation is

$$\mathbf{X}^T \mathbf{X} \mathbf{w} = \mathbf{X}^T \mathbf{y}$$

where

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 4 & 0 & 4 \\ 0 & 1 & -1 \\ 4 & -1 & 5 \end{bmatrix}, \quad \mathbf{X}^T \mathbf{y} = \begin{bmatrix} 6 \\ 2 \\ 4 \end{bmatrix}$$

There are infinitely many solutions given by

$$\mathbf{w} = (1.5, 2, 0) + \alpha(-1, 1, 1), \quad \alpha \in \mathbb{R}.$$

General unconstrained QP

Minimize quadratic function with $\mathbf{Q} \in \mathbb{R}^{n \times n}$ s.t. $\mathbf{Q} \succeq \mathbf{O}$,

$$\min_{\mathbf{x}} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$$

By the first-order condition, $\mathbf{x}^*$ is optimal iff it is a solution to

$$\nabla f(\mathbf{x}) = \mathbf{Q} \mathbf{x} + \mathbf{b} = \mathbf{0}$$

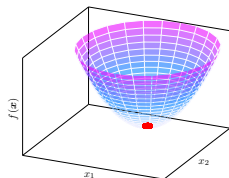
Case I. $\mathbf{Q} \succ \mathbf{O}$: unique solution $\mathbf{x}^* = -\mathbf{Q}^{-1} \mathbf{b}$

Case II. $\det \mathbf{Q} = 0$ and $\mathbf{b} \in \text{Range}(\mathbf{Q})$ ¹: infinitely many solutions (why?)

Case III. $\det \mathbf{Q} = 0$ and $\mathbf{b} \notin \text{Range}(\mathbf{Q})$: no solution, and $f^* = -\infty$.

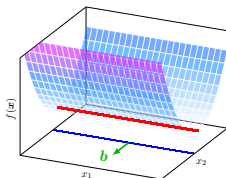
¹This is equivalent to $-\mathbf{b} \in \text{Range}(\mathbf{Q})$, the latter being just another way to say $\mathbf{Q} \mathbf{x} + \mathbf{b} = \mathbf{0}$ has a solution.

General unconstrained QP (cont'd)



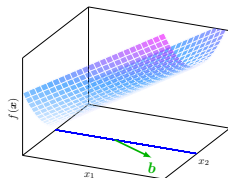
$$\mathbf{Q} \succ \mathbf{O}$$

unique solution



$$\mathbf{b} \perp \text{Null}(\mathbf{Q}) \neq \{\mathbf{0}\}$$

infinitely many solutions



$$\mathbf{b} \not\perp \text{Null}(\mathbf{Q}) \neq \{\mathbf{0}\}$$

no solution

To understand why $f^* = -\infty$ in case III, recall $\text{Null}(\mathbf{A})^\perp = \text{Range}(\mathbf{A}^T)$. Since $\mathbf{Q}^T = \mathbf{Q}$,

$$\mathbf{b} \in \text{Range}(\mathbf{Q}) = \text{Range}(\mathbf{Q}^T) \iff \mathbf{b} \in \text{Null}(\mathbf{Q})^\perp \iff \mathbf{b} \perp \text{Null}(\mathbf{Q})$$

Since $\mathbf{b} \not\perp \text{Null}(\mathbf{Q})$, there exists a nonzero $\mathbf{x}_0 \in \text{Null}(\mathbf{Q})$ s.t. $\mathbf{b}^T \mathbf{x}_0 \neq 0$,

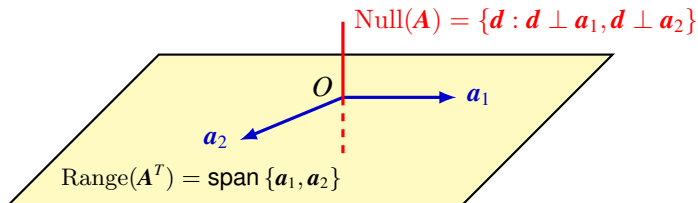
$$f(t\mathbf{x}_0) = t\mathbf{b}^T \mathbf{x}_0 + c$$

Letting $t \rightarrow -\infty$ if $\mathbf{b}^T \mathbf{x}_0 > 0$ and $t \rightarrow +\infty$ if $\mathbf{b}^T \mathbf{x}_0 < 0$, $f(t\mathbf{x}_0) \rightarrow -\infty$.

Appendix

Lemma. $\text{Null}(\mathbf{A})^\perp = \text{Range}(\mathbf{A}^T)$, where $\text{Null}(\mathbf{A})^\perp$ is the orthogonal complement of $\text{Null}(\mathbf{A})$, i.e.

$$\mathbf{x} \in \text{Null}(\mathbf{A})^\perp \iff \mathbf{x} \perp \mathbf{d}, \quad \forall \mathbf{d} \in \text{Null}(\mathbf{A})$$



Proof. Show $\text{Range}(\mathbf{A}^T) \subset \text{Null}(\mathbf{A})^\perp$ is a subspace with the same dimension, so $\text{Range}(\mathbf{A}^T) = \text{Null}(\mathbf{A})^\perp$.

- $\mathbf{x} \in \text{Range}(\mathbf{A}^T) \implies \mathbf{x} = \mathbf{A}^T \mathbf{z}$ for some $\mathbf{z}$
- $\forall \mathbf{d} \in \text{Null}(\mathbf{A}), \mathbf{x}^T \mathbf{d} = \mathbf{z}^T \mathbf{A} \mathbf{d} = \mathbf{z}^T \mathbf{0} = 0$, i.e. $\mathbf{x} \perp \mathbf{d}$, so $\mathbf{x} \in \text{Null}(\mathbf{A})^\perp$.
- $\dim \text{Range}(\mathbf{A}^T) = \text{rank } \mathbf{A} = n - \dim \text{Null}(\mathbf{A}) = \dim \text{Null}(\mathbf{A})^\perp$

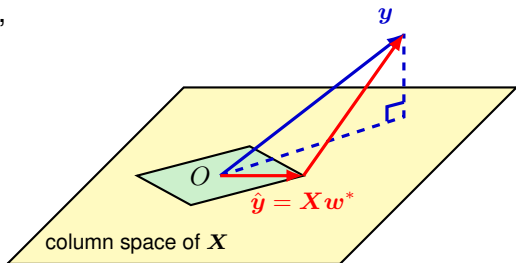
Example: Lasso

Lasso (Least Absolute Shrinkage and Selection Operator)

Given $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times p}$, $t > 0$,

$$\begin{aligned} \min_{\mathbf{w}} \quad & \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 \\ \text{s. t.} \quad & \|\mathbf{w}\|_1 \leq t \end{aligned}$$

- convex problem? yes
- QP? no, but can be converted to QP
- optimal solution exists? yes
 - ▶ compact feasible set
- optimal solution unique?
 - ▶ yes if $n \geq p$ and $\mathbf{X}$ has full column rank ($\mathbf{X}^T \mathbf{X} \succ \mathbf{O}$, strictly convex)
 - ▶ no in general, e.g. $p > n$ and t is large enough for unconstrained optima to be feasible

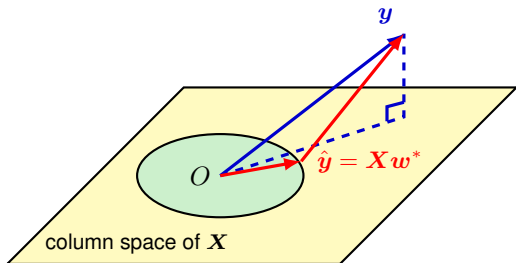


Example: Ridge regression

Given $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times p}$, $t > 0$,

$$\begin{aligned} \min_{\mathbf{w}} \quad & \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 \\ \text{s. t.} \quad & \|\mathbf{w}\|_2^2 \leq t \end{aligned}$$

- convex problem? yes
- QCQP? yes



- optimal solution exists? yes
 - ▶ compact feasible set
- optimal solution unique?
 - ▶ yes if $n \geq p$ and $\mathbf{X}$ has full column rank ($\mathbf{X}^T \mathbf{X} \succ \mathbf{O}$, strictly convex)
 - ▶ no in general

Example: SVM

Linearly separable case

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s. t.} \quad & y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \quad i = 1, 2, \dots, m \end{aligned}$$

Soft margin SVM

$$\begin{aligned} \min_{\mathbf{w}, b, \boldsymbol{\xi}} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{i=1}^m \xi_i \\ \text{s. t.} \quad & y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i, \quad i = 1, 2, \dots, m \\ & \boldsymbol{\xi} \geq \mathbf{0} \end{aligned}$$

Equivalent unconstrained form

$$\min_{\mathbf{w}, b} \quad \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{i=1}^n (1 - y_i b - y_i \mathbf{x}_i^T \mathbf{w})^+$$

Outline

- Quadratic program and quadratically constrained QP
- Geometric program

Geometric program

A **monomial** is a function $f : \mathbb{R}_{++}^n = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} > \mathbf{0}\} \rightarrow \mathbb{R}$ of the form

$$f(\mathbf{x}) = \gamma x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$$

for $\gamma > 0$, $a_1, \dots, a_n \in \mathbb{R}$. A **posynomial** is a sum of monomials,

$$f(\mathbf{x}) = \sum_{k=1}^p \gamma_k x_1^{a_{k1}} x_2^{a_{k2}} \cdots x_n^{a_{kn}}$$

A **geometric program** (GP) is an optimization problem of the form

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s. t.} \quad & g_i(\mathbf{x}) \leq 1, \quad i = 1, \dots, m \\ & h_j(\mathbf{x}) = 1, \quad j = 1, \dots, r \end{aligned}$$

where $f, g_i, i = 1, \dots, m$ are posynomials and $h_j, j = 1, \dots, r$ are monomials. The constraint $\mathbf{x} > \mathbf{0}$ is implicit.

Geometric program (cont'd)

GP is nonconvex (why?)

$$\begin{aligned} \min_{\mathbf{x}} \quad & \sum_{k=1}^{p_0} \gamma_{0k} x_1^{a_{0k1}} x_2^{a_{0k2}} \cdots x_n^{a_{0kn}} \\ \text{s. t.} \quad & \sum_{k=1}^{p_i} \gamma_{ik} x_1^{a_{ik1}} x_2^{a_{ik2}} \cdots x_n^{a_{ikn}} \leq 1, \quad i = 1, \dots, m \\ & \eta_j x_1^{c_{j1}} x_2^{c_{j2}} \cdots x_n^{c_{jn}} = 1, \quad j = 1, \dots, r \end{aligned}$$

By $y_i = \log x_i$, $b_{ik} = \log \gamma_{ik}$, $d_j = \log \eta_j$, GP can be formulated as

$$\begin{aligned} \min_{\mathbf{y}} \quad & \log \left(\sum_{k=1}^{p_0} e^{\mathbf{a}_{0k}^T \mathbf{y} + b_{0k}} \right) \\ \text{s. t.} \quad & \log \left(\sum_{k=1}^{p_i} e^{\mathbf{a}_{ik}^T \mathbf{y} + b_{ik}} \right) \leq 0, \quad i = 1, \dots, m \\ & \mathbf{c}_j^T \mathbf{y} + d_j = 0, \quad j = 1, \dots, r \end{aligned}$$

This is convex by the convexity of log-sum-exp (soft max) functions

Geometric program (cont'd)

Example. Let $u = \log x$, $v = \log y$, $w = \log z$.

$$\begin{aligned} \min_{x,y,z>0} \quad & x^{-1}y + xz \\ \text{s. t.} \quad & 2x^{-1} \leq 1 \\ & \frac{1}{3}x \leq 1 \\ & x^2y^{-1/2} + 3y^{1/2}z^{-1} \leq 1 \\ & x^2y^{-1}z^{-2} = 1 \end{aligned}$$

is equivalent to

$$\begin{aligned} \min_{u,v,w} \quad & \log(e^{-u+v} + e^{u+w}) \\ \text{s. t.} \quad & \log 2 - u \leq 0 \\ & -\log 3 + u \leq 0 \\ & \log(e^{2u-\frac{1}{2}v} + e^{\log 3 + \frac{1}{2}v-w}) \leq 0 \\ & 2u - v - w = 0 \end{aligned}$$